

Features of AEROSecure

The system includes authentication, role-based access control, task management, and security monitoring.

There are 3 user roles:

1. Admin
2. Manager
3. Employee

AUTHENTICATION & SECURITY:

- Users log in with username and password (bcrypt hashing)
- System captures login details: IP address, device, location, and time
- Logs all activities (login, failed login, suspicious activity)

Security Rules Implemented

1. First Threshold – 3 Failed Attempts

- After 3 consecutive failed login attempts:
 - User account is temporarily locked for 1 minute
 - A warning email alert is sent to the user/admin
 - Further login attempts are restricted during this period

2. Second Threshold – 5 Failed Attempts

- After 5 consecutive failed login attempts:
 - User account is locked for 5 minutes
 - A suspicious activity alert email is triggered
 - System flags the activity as potential security threat
 - Access is strictly blocked during lock period
- Sends email alerts (using Mailtrap SMTP) for:
 - Successful login

- multiple Failed login attempts
- Suspicious activity
- creation/deletion of employee
- new task assigned to a employee, with deadline
- system send email of username and password of new profile created by admin

ADMIN FEATURES:

- Create new employees (username + password)
- Delete existing employees
- View all users
- Assign tasks to employees
- View all tasks and their status
- View system logs (security monitoring)

MANAGER FEATURES:

- Assign tasks to employees
- Monitor task progress
- Filter and search tasks
- View employee performance

EMPLOYEE FEATURES:

- View assigned tasks
- Mark tasks as completed (checklist system)
- Upload files as proof of task completion
- View task deadlines

TASK MANAGEMENT:

- Admin/Manager assigns tasks to employees

- Each task includes:
 - Task name
 - Assigned user
 - Status (pending/completed)
 - Deadline
 - Optional file upload
- Employees can mark tasks as completed

EMAIL SYSTEM:

- Email alerts are sent using Mailtrap (SMTP)
- Emails are triggered for:
 - Login activity
 - Task assignment
 - Security alerts

SEARCH & FILTER:

- Admin/Manager can:
 - Search users
 - Filter tasks (pending/completed)
 - Sort by deadline

ADDITIONAL FEATURES:

- Task deadline system (highlight overdue tasks)
- Role-Based Access Control (RBAC)
- Activity logging system (audit trail)
- File upload system for tasks
- UI improvements (clean dashboard, tables, colors, responsive design)

UI:

- Separate dashboards for Admin, Manager, and Employee
- Clean and user-friendly interface
- Tables for logs, tasks, and users
- Status indicators (green = completed, red = pending)

GOAL:

The system aims to improve airport operational efficiency and security by combining task management, real-time monitoring, role-based access, and automated alerts.

ADDITIONAL FEATURES

Live Dashboard (Real-Time Feel)

Active users

Tasks in progress

Task Escalation System

If task not completed before deadline:

Automatically:

-Notify manager

-Increase priority

-Send alert

Digital Signature / Approval

Manager must **approve completed task**

Adds:

“Verified by Manager”

Multi-Step Login (2-Level Security)

After password:

Send OTP (simulated via email)

User must verify

Login Time Restriction

Employees can only login:

Between 8 AM – 8 PM (working hours) (also depends on employee working hours, day or night shift)

Outside this:

- Block login

- Send alert

Daily Summary Email

At end of day:

Admin gets email:

- Tasks completed

- Pending tasks

- Alerts